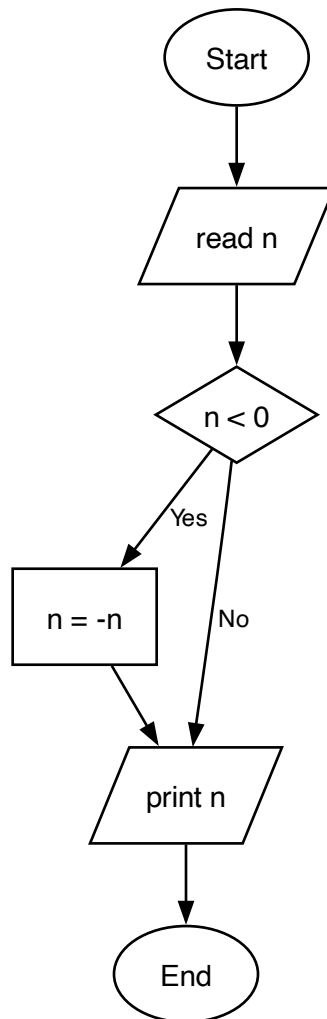


Name: \_\_\_\_\_

Date: \_\_\_\_\_

**Problem 1.**

(i) For each input below, follow the flowchart and write the output.

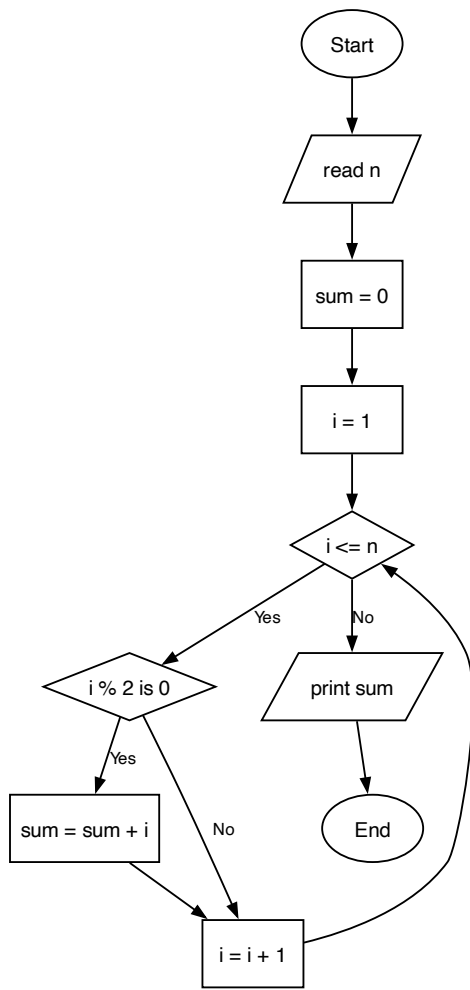
| input $n$ | output |
|-----------|--------|
| 7         |        |
| -4        |        |
| 0         |        |
| -19       |        |

Trace this algorithm for  $n = 4$ , and complete the trace table.

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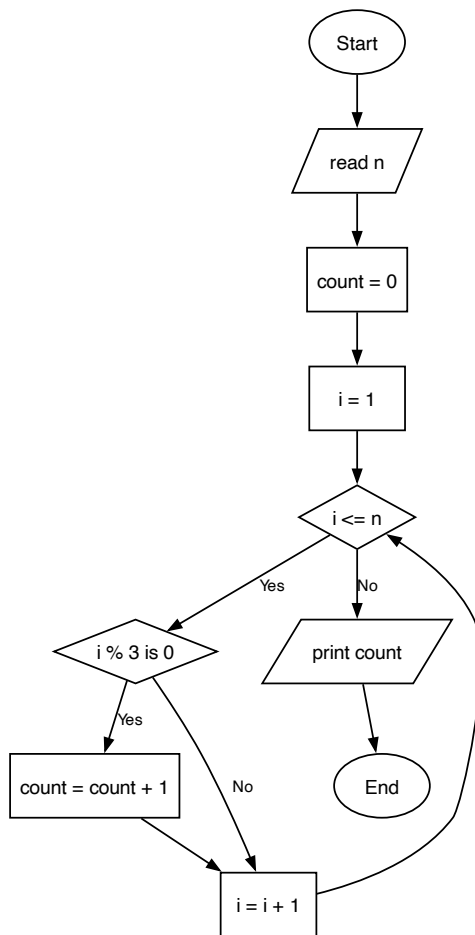
### Problem 3.

Trace this algorithm for  $n = 5$ , and complete the trace table.

[illegible]

### Problem 4.

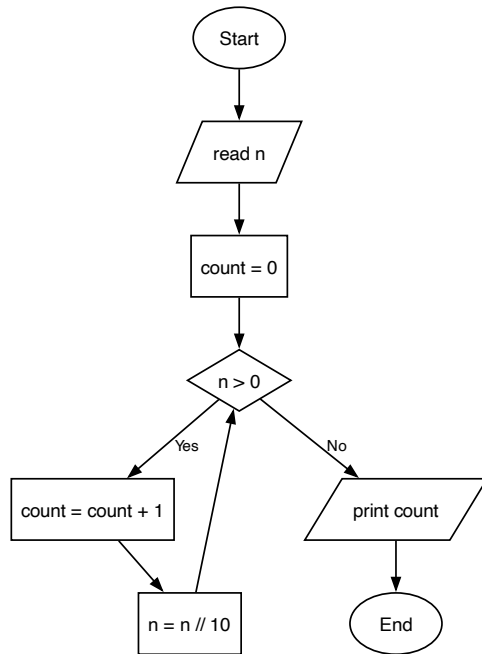
Trace this algorithm for  $n = 7$ , and complete the trace table.

[illegible]

### Problem 5.

Trace this algorithm for  $n = 472$ , and complete the trace table.

*Note:  $n // 10$  means whole-number division — divide and drop the remainder. For example,  $472 // 10 = 47$  (not  $47.2$ ).*



| n | count | output |
|---|-------|--------|
|   |       |        |
|   |       |        |
|   |       |        |
|   |       |        |
|   |       |        |
|   |       |        |
|   |       |        |
|   |       |        |
|   |       |        |
|   |       |        |

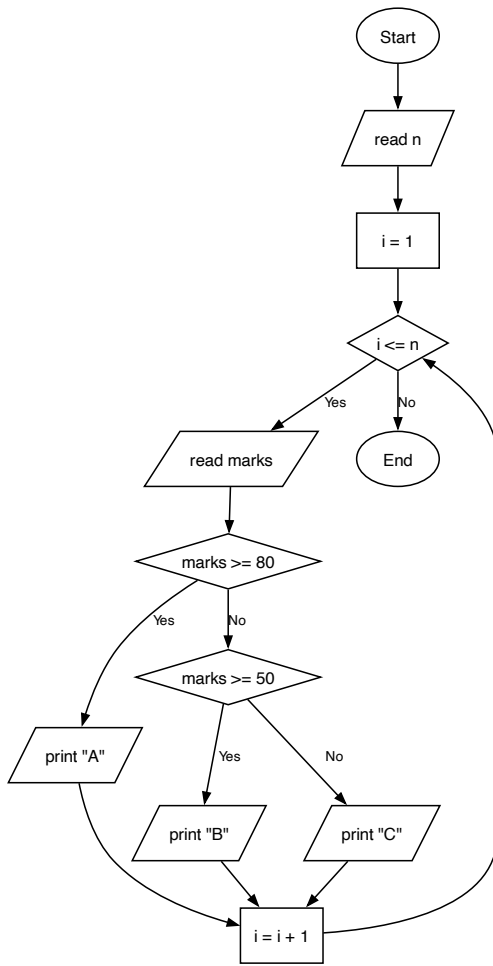
### Problem 6.

*Ask the user for a number  $n$ . Then go through the numbers 1, 2, 3, ... up to  $n$ , one by one: if a number is divisible by 3, print “buzz”; otherwise print the number itself.*

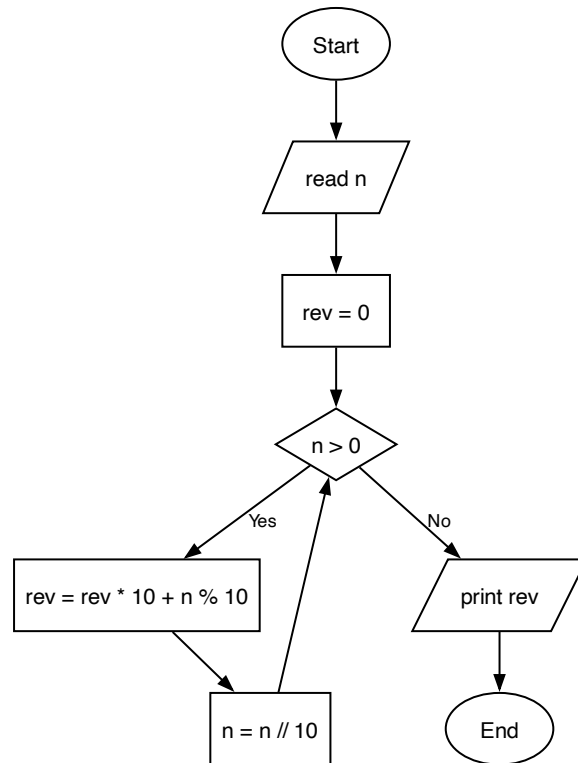
Draw a flowchart for this algorithm:

### Problem 7.

Trace this algorithm for  $n = 3$ , where the values of *marks* entered are 55, 82, 40 (in that order), and complete the trace table.

[illegible]

### Problem 8.



(i) For each input below, follow the flowchart and write the output.

*Note:  $n \% 10$  gives the remainder after dividing by 10 — the last digit of  $n$ . For example,  $472 \% 10 = 2$ .*

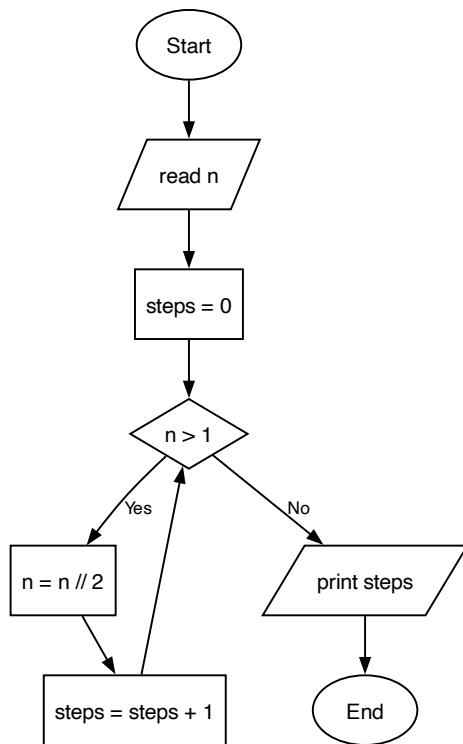
| input $n$ | output |
|-----------|--------|
| 123       |        |
| 47        |        |
| 5         |        |
| 470       |        |

(ii) Can you tell what this algorithm is doing?

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### Problem 9.

(i) Trace this algorithm for  $n = 16$ , and complete the trace table.

[illegible]

(ii) For each input below, follow the flowchart and write the output.

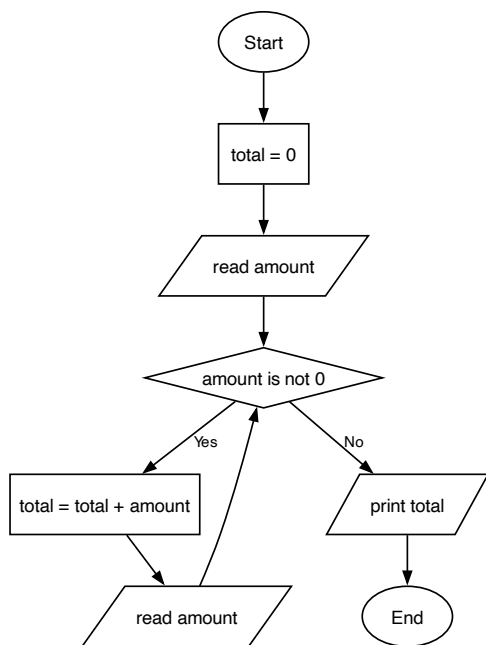
| input $n$ | output |
|-----------|--------|
| 8         |        |
| 16        |        |
| 100       |        |
| 1000      |        |

(iii)  $n$  grew from 8 all the way to 1000 — but look how little the output grew! Can you explain why?

### Problem 10.

*At the end of the day, a shopkeeper adds up the money received from each customer. She enters the amounts one by one, and enters 0 when there are no more. The program then shows the total. Notice that the algorithm cannot know in advance how many customers there were!*

Trace this algorithm, where the values of *amount* entered are 120, 75, 200, 0 (in that order), and complete the trace table.



| total | amount | output |
|-------|--------|--------|
|       |        |        |
|       |        |        |
|       |        |        |
|       |        |        |
|       |        |        |
|       |        |        |
|       |        |        |
|       |        |        |
|       |        |        |
|       |        |        |

### Problem 11.

*After an exam, the teacher enters each student's marks one by one, and enters -1 when there are no more. The program should then print three things: how many students took the exam, their average marks, and how many of them passed (40 or more). (Why is -1 a better "no more" signal here than 0?)*

Draw a flowchart for this algorithm: